

Homework Assignment 5

1. A turbojet engine has TSFC = 1.1 h^{-1} . What is its specific impulse? *From the units, you should recognize that the weight based TSFC has been given.* {Ans.: 3273 s}
2. A turbofan engine has core air mass flow rate of 100 kg/s and bypass air mass flow rate of 30 kg/s. What is the bypass ratio? Would this engine be classified as a LBTF engine or a HBTF engine? Also, what is the air mass flow rate entering the engine?
3. What is the value of bypass ratio of a turbojet engine?
4. Reconsider the HBTF example of class (bypass ratio 6, $\tau = 103100 \text{ N}$). Calculate the propulsive efficiency of the engine. {Ans.: 74.45%}
5. A twin-engine airplane has wing area of 93 m^2 and drag coefficient of 0.023. Each of the turbofan engines has TSFC of 0.63 h^{-1} during steady, level flight at $M = 0.84$ at altitude of 10000 m ($T_a = 223 \text{ K}$, $p_a = 26.5 \text{ kPa}$, $\gamma = 1.4$, $R = 287 \text{ J/kg-K}$). Calculate (a) the thrust produced by each engine, (b) the fuel flow rate into each engine in kg/h and (c) the overall efficiency of the engine. $Q_R = 43000 \text{ kJ/kg}$. {Ans.: 13999 N, 899 kg/h, $\eta_0 = 32.78\%$ }
6. Using the Breguet range equation, calculate the range of the airplane of Problem 5 for a cruise fuel load of 5000 kg. The mass (which includes fuel) of the airplane at the start of cruise is 50000 kg. {Ans.: 2652 km}